CHAMATH DE SILVA

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EDUCATION

Bachelor of Applied Science - Co-op (BASc) — Sept 2023 – Present
Level 3 - Honours Computer Science — McMaster University Hamilton, Ontario
Relevant Coursework: Operating Systems | Data Structures and Algorithms | Software Testing | Networks and Security

TECHNICAL SKILLS

Languages & AI: Python (Expert), SQL (Advanced), Java, C++, R; Supervised/Unsupervised Learning, Neural Networks (CNN/RNN/Transformers), Ensemble Methods (XGBoost/Random Forest), Hyperparameter Tuning.
Data Science Stack: NumPy, pandas, Scikit-Learn, TensorFlow/PyTorch, Matplotlib, Seaborn, StatCan/yfinance APIs.
Technical Competencies: Statistical Modeling, ETL/ELT Pipeline Design, Data Visualization, BI Reporting (Tableau/Domo).

PROJECTS

AI/ML Stock Prediction Application *Python | Scikit-Learn | yfinance | Ensemble Learning* — 2026

- Built an ensemble inference engine using Random Forest and Gradient Boosting to predict price movements, achieving high consistency through automated feature engineering of technical indicators (RSI, MACD).
- Engineered a modular data pipeline to process 6,000+ historical records, successfully decoupling offline training from online inference to enable rapid model iteration and deployment.
- Implemented a modular "Offline Training / Online Inference" architecture to ensure models can be retrained and redeployed without disrupting the live Streamlit dashboard.

Volatility Risk Engine *Python | Statistical Analysis | Matplotlib | Data Engineering* — 2025 - 2026

- Implemented a real-time statistical audit system utilizing Z-Score analysis ($|Z| > 2$) to isolate price anomalies and detect fraudulent listings in unstructured hardware market data.
- Designed a weighted risk matrix combining price variance and seller trust scores into a 4-tier classification model, visualized through automated performance-trend reports and distribution plots.
- Automated PDF/PNG dashboard generation using Matplotlib and Seaborn for professional visual audit trails.

CanAfford: AI-Driven Tenant Platform *React | Google Gemini | Backboard API | LLM Pipelines* — 2026

- Engineered a Legal Audit Agent utilizing a constrained LLM pipeline to parse complex lease documents and autonomously flag Ontario Residential Tenancies Act (RTA) violations with high precision.
- Developed a "True Cost" dashboard that synthesizes live inflation and transit data, utilizing bi-directional state management to model and predict individual survival costs.
- Built a Bi-Directional State Management system via Backboard API to autonomously sync revealed user preferences into a dynamic behavioral model for personalized risk assessment.

WORK EXPERIENCE

Software Engineer Intern | *Callidora Trading Company Ltd - Guelph, ON* — May 2026 – Present

- Engineering scalable ETL/ELT data pipelines and backend services in Python and SQL to support high-concurrency machine learning and large-scale analytics workflows.
- Optimizing data ingestion and storage within Google Cloud Platform (GCP) to ensure high system performance, reliability, and automated validation for production datasets.
- Collaborating with cross-functional teams to translate business requirements into technical designs focused on horizontal scalability and long-term system maintainability.

CERTIFICATIONS

Machine Learning & Artificial Intelligence Certificate | *University of Waterloo (SCS)* — 2025
CS109x: Introduction to Data Science with Python | *Harvardx* — Aug 2025